

DIGITAL IMAGE PROCESSING MINI PROJECT REPORT

# SmartFruit Vision

Restoration and Segmentation of Fruit Images for  
Automated Defect Detection

*Digital Image Processing — Mini Project*

Course Code: 265DSEUBC302B

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Git hub link: <https://github.com/AneenaBinoy/SmartFruit-Vision.git>

## 1. Title

SmartFruit Vision: Restoration and Defect Segmentation of Fruit Images Using Classical Image Processing Techniques.

## 2. Problem Statement

The project aims to identify and analyze visible surface damage in fruit images using Digital Image Processing.

Manual inspection of fruit quality is slow, subjective, and difficult to scale in retail and agricultural settings. This project addresses the problem of automatically detecting and quantifying visible surface defects (discoloration, spoilage) on fruit images, while also demonstrating recovery of fruit images degraded by noise. The system takes a fruit image as input, restores it if corrupted by noise, isolates the fruit from the background, and segments and quantifies the visibly defective region on its surface.

## 3. Objectives

- To study and apply image restoration techniques (mean, median, and Gaussian filtering) for removing Gaussian noise from fruit images.
- To evaluate restoration quality quantitatively using MSE, PSNR, and SSIM.
- To enhance restored images using histogram equalization and CLAHE.
- To segment the fruit region from the background using GrabCut and morphological post-processing.
- To segment surface defects on the fruit using K-Means color clustering in the LAB color space.
- To quantify defect severity as a percentage of fruit surface area and classify severity level.
- To analyze the spatial distribution of defects across the fruit surface (top / middle / bottom regions).

## 4. Dataset / Image Description

**Dataset name:** SmartFruit\_Vision (fresh and rotten fruit image collection)

**Categories:** freshapples, freshbanana, freshoranges, rottenapples, rottenbanana, rottenoranges

**Image characteristics:** Colour (RGB) images of individual fruits, variable resolution, JPG/PNG/BMP format

**Number of images used:** [Fill in from the 'Checking dataset...' output — total image count per category and overall]

**Preprocessing applied:** Conversion to RGB/NumPy array; synthetic Gaussian noise ( $\sigma = 25$ ) added to a sample image to simulate a degraded/noisy acquisition scenario for the restoration stage

Dataset limitations: images are taken against varied backgrounds and lighting, and only one sample per class was carried through the full restoration and defect-analysis pipeline in this notebook, so results are illustrative rather than dataset-wide.



## 5. Methodology

The project follows two complementary pipelines: a **restoration pipeline** applied to a noisy fruit image and a **segmentation pipeline** applied to fresh and rotten fruit images for visible surface defect analysis.

### 5.1 Restoration Pipeline

**Input Image → Add Gaussian Noise → Mean / Median / Gaussian Filtering → Quantitative Evaluation (MSE, PSNR, SSIM) → Histogram Equalization / CLAHE Enhancement → Comparison**

- **Input Image:** Rotten Apple 20 is selected as the main restoration image.
- **Add Gaussian Noise:** Artificial Gaussian noise is added to simulate image degradation.
- **Filtering:** Mean, Median and Gaussian filters are applied to reduce the noise.
- **Evaluation:** MSE, PSNR and SSIM are calculated to compare the filtering results.
- **Enhancement:** Histogram Equalization and CLAHE are applied to improve image contrast.
- **Comparison:** The filtering methods are compared using visual results and PSNR values.

### 5.2 Segmentation and Defect Analysis Pipeline

**Input Fruit Image → GrabCut Foreground Segmentation → Morphological Cleaning → Largest-Contour Fruit Mask → Fruit Extraction → K-Means Clustering (LAB Colour Space, k = 5) → Defect Cluster Selection → Morphological Cleaning of Defect Mask → Defect Boundary Highlighting → Defect Area / Severity / Regional Distribution Analysis → Automated Quality Report**

- **Input Fruit Image:** Fresh Apple 7 and Rotten Apple 20 are selected.
- **GrabCut:** The fruit is separated from the background.
- **Morphological Cleaning:** Opening and closing operations improve the fruit mask.

- **Largest Contour:** The main fruit region is retained.
- **Fruit Extraction:** The segmented fruit is separated from the background.
- **K-Means:** The fruit is divided into five colour clusters in LAB colour space.
- **Defect Selection:** Cluster 0 is selected as the visible discoloration/defect region for Rotten Apple 20.
- **Morphological Cleaning:** Small unwanted regions are reduced and the defect mask is refined.
- **Defect Highlighting:** The detected defect boundary is highlighted on the original image.
- **Analysis:** Visible defect percentage, severity level and regional distribution are calculated.
- **Quality Report:** The final analysis is presented as an automated report.

## 6. Implementation

**Programming language:** Python (Google Colab)

**Libraries used:** OpenCV (cv2), NumPy, Matplotlib, PIL/Pillow, scikit-image (skimage.metrics, skimage.exposure), scikit-learn (KMeans)

### 6.1 Restoration techniques

- Gaussian noise simulation: additive noise with mean 0, standard deviation 25
- Mean filter: cv2.blur, kernel 5×5
- Median filter: cv2.medianBlur, kernel size 5
- Gaussian filter: cv2.GaussianBlur, kernel 5×5
- Enhancement: histogram equalization and CLAHE (clip\_limit = 0.03) via skimage.exposure

### 6.2 Segmentation techniques

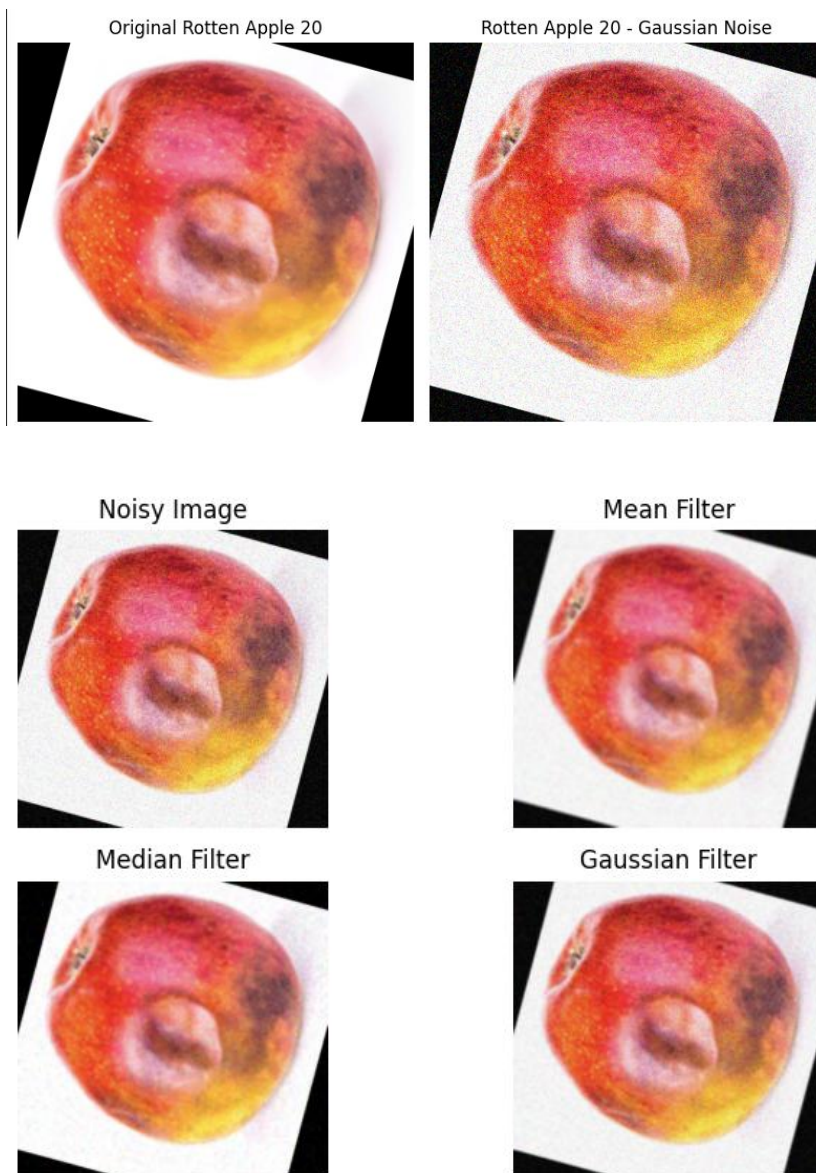
- GrabCut (cv2.grabCut) with a rectangular initialization covering ~90% of the image, refined over 5 iterations
- Morphological opening and closing (7×7 kernel) to clean the foreground mask
- Largest external contour selection to isolate the fruit object

- K-Means clustering ( $k = 5$ ) on LAB pixel values restricted to the fruit mask, to separate colour/defect regions
- Manual selection of the cluster corresponding to the visible defect/discoloration region
- Morphological opening ( $5 \times 5$ ) and closing ( $11 \times 11$ ) to clean the defect mask
- Contour-based boundary drawing to highlight detected defects on the original image

## 7. Results

### 7.1 Restoration Results

A Gaussian-noise-corrupted image was restored using three spatial filters.



## RESTORATION RESULTS

### Mean Filter

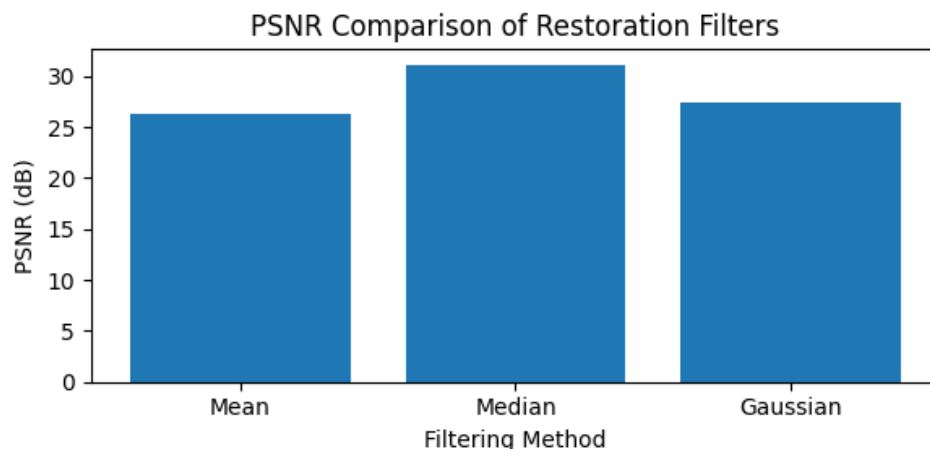
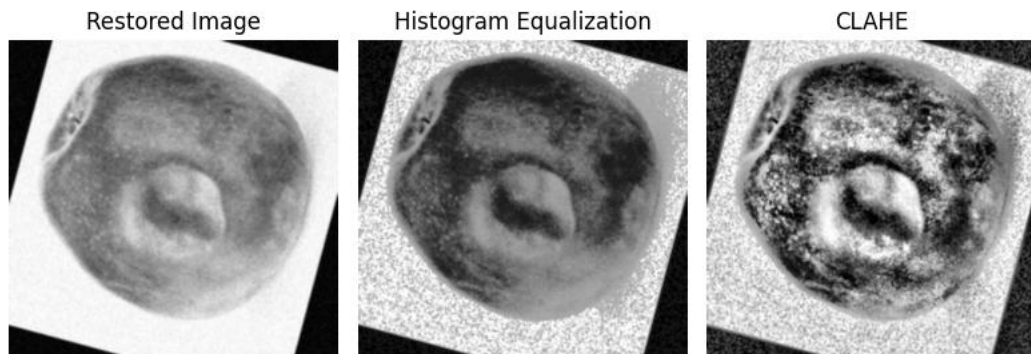
MSE : 152.8104  
PSNR : 26.2893 dB  
SSIM : 0.7844

### Median Filter

MSE : 50.6959  
PSNR : 31.0811 dB  
SSIM : 0.7757

### Gaussian Filter

MSE : 118.5552  
PSNR : 27.3916 dB  
SSIM : 0.7352



## 7.2 Segmentation and Defect Detection Results

Fresh Apple 7



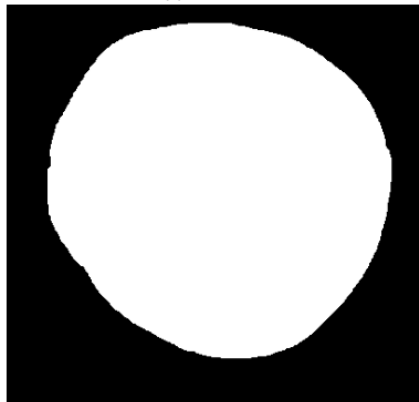
Rotten Apple 20



Fresh Apple 7 - Fruit Mask



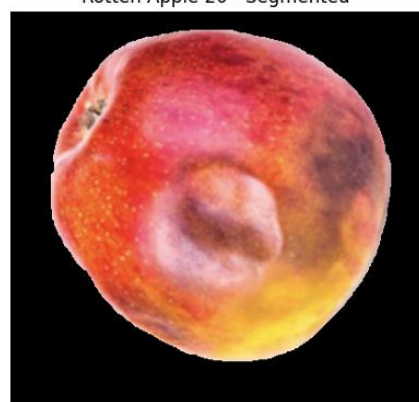
Rotten Apple 20 - Fruit Mask



Fresh Apple 7 - Segmented



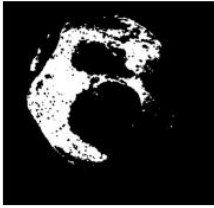
Rotten Apple 20 - Segmented



Rotten Apple 20



Cluster 0



Cluster 1



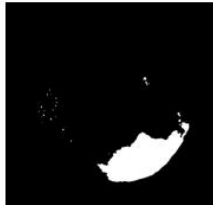
Cluster 2



Cluster 3



Cluster 4



Rotten Apple 20 - Cleaned Defect Mask



Visible defect regions highlighted.

Rotten Apple 20 - Original



Cluster 3 - Defect Mask



Detected Defect Regions



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=====
DEFECT DISTRIBUTION
=====
Top Region      : 5857 pixels
Middle Region   : 7741 pixels
Bottom Region   : 221 pixels
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Most Damaged Region : Middle
=====

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SMARTFRUIT VISION - QUALITY REPORT
=====
Fruit Sample      : Rotten Apple 20
Visible Defect Area : 16.17%
Defect Severity    : Moderate
Most Damaged Region : Middle
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Assessment         : Moderate visible surface damage detected.
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Note: This assessment is based on visible
surface regions detected using image processing.
It is not a food-safety or spoilage diagnosis.
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## 8. Limitations

- The restoration pipeline was demonstrated on a single noisy image rather than a batch across the dataset, so PSNR/SSIM improvements are illustrative and not averaged over multiple samples.
- The segmentation and defect-analysis pipeline was likewise run on one fresh and one rotten sample; results have not been aggregated across the full dataset.
- The defect cluster (Cluster 3) was selected manually by visual inspection rather than through an automatic rule, so the pipeline does not yet generalize unattended to new images.
- No pixel-level ground-truth defect mask was available, so segmentation quality could not be evaluated quantitatively (e.g., IoU, Dice coefficient) — only qualitative visual assessment and area/percentage measures were used.
- GrabCut initialization assumes the fruit is roughly centered and does not touch the image border; performance may degrade on differently framed images.
- The system detects visible surface discoloration only and does not diagnose actual spoilage, ripeness, or food safety.

## 9. Future Scope

- Extend the pipeline to run automatically across the full dataset and report aggregate statistics per category.
- Replace manual defect-cluster selection with an automatic rule (e.g., distance of cluster centroid from a reference healthy-fruit colour in LAB space).
- Collect or annotate ground-truth defect masks for a subset of images to enable quantitative segmentation evaluation (IoU, Dice).
- Compare classical segmentation against deep-learning-based approaches (e.g., a CNN or U-Net defect segmentation model).
- Deploy the system as a real-time web or mobile application for practical fruit-quality screening.
- Test robustness to real (non-synthetic) noise and varied lighting/background conditions.

## 10. Conclusion

This project developed SmartFruit Vision, an image processing pipeline that combines restoration and segmentation to address fruit quality assessment. Gaussian noise was removed using mean, median, and Gaussian filters, evaluated quantitatively using MSE, PSNR, and SSIM, and further improved using histogram equalization and CLAHE. Fruit regions were isolated from the background using GrabCut with morphological refinement, and surface defects were segmented using K-Means colour clustering in the LAB space, followed by morphological cleaning and contour-based visualization. The system quantified defect severity as a percentage of fruit surface area, classified severity level, and analyzed the spatial distribution of damage. The project demonstrates that classical DIP techniques, applied together, can build a coherent, interpretable, and reasonably effective fruit-defect-detection system, while also highlighting clear directions — larger-scale evaluation, automatic defect-cluster selection, and ground-truth-based metrics — for future improvement.